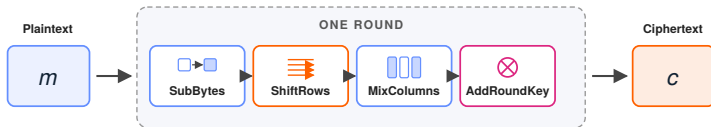




Advanced Encryption Schemes

Cecilia Boschini and Kristina Hostáková

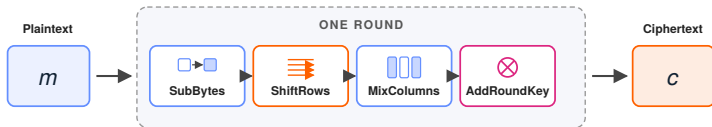
AES: talking about symmetric encryption... right?



AES — Advanced Encryption Standard

Rijndael, Daemen & Rijmen · FIPS 197 (2001) · a symmetric block cipher

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Wrong AES. This course: **Advanced Encryption Schemes.**

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- Goal:
Study techniques to obtain provably secure (mainly *public key*) encryption schemes with advanced properties.

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Study techniques to obtain provably secure (mainly *public key*) encryption schemes with advanced properties.
- “Restarted” lecture
 - Formerly thought by Romain Gay
 - Goal is the same, but...
 - ...content significantly different!

Organization - Lecture

- Lectures: Thursdays, 12:15 - 14:00, HG D3.2
- Lecturers:
 - Kristina Hostáková (first part of the course)
 - Cecilia Boschini (second part of the course)
- Oral exam (15mins) in January/February

Organizational - Learning Material

- Main source: Slides from lectures (uploaded in Moodle)
- Additional sources:
 - Book of Nigel Smart, “Cryptography made simple” (see [here](#))
 - Some lectures might have additional reading materials

Organization - Exercise sessions

- Bi-weekly, Thursdays 16:15 - 18:00, CHN E46
 - First session: 24.9.2026
 - Exercise sheet - published in Moodle a week before
- TAs: Rune Fiedler, Xiaoqi Duan

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Organization - Contacts

- Moodle Forum!
 - About the lecture or exercise content/logistics.
 - Might be relevant for all students.
 - Interaction welcome!

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- Teaching Assistants
 - rune.fiedler@inf.ethz.ch, xiaoqi.duan@inf.ethz.ch
 - Questions about the exercise sessions or the assignment.
- Lecturers:
 - kristina.hostakova@inf.ethz.ch, cecilia.boschini@inf.ethz.ch

Outline

Logistics

Motivation

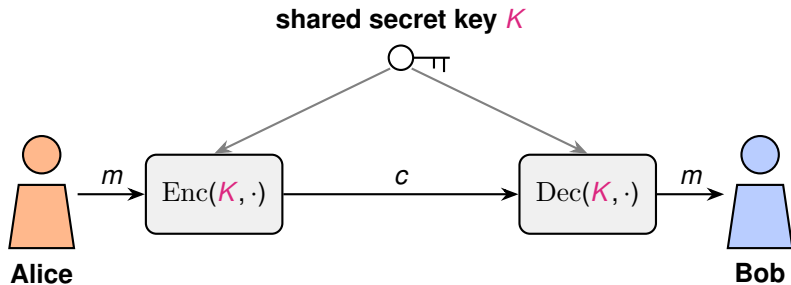
Syntax and Definitions

Semantic Security

Game-based security

Semantic security equivalent to IND-CPA

Symmetric encryption

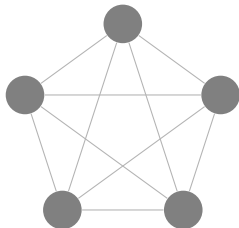


Same key K both locks and unlocks the message.

Fast and simple — *as long as* Alice and Bob already share K .

Symmetric encryption: Key Distribution Problem

- Fast and well understood.
- **But:** Alice and Bob need a *shared secret key*.
- How do they agree on it?
 - Meet in person? Not scalable.
 - Courier? Slow, expensive, needs trust.
- With n pairwise communicating parties: $\binom{n}{2} = O(n^2)$ keys to distribute and manage.



Pairwise shared keys: quadratic
blow-up

The idea: separate the keys

- Diffie–Hellman (1976) and Rivest–Shamir–Adleman (1977):
what if encryption and decryption used *different* keys?

The idea: separate the keys

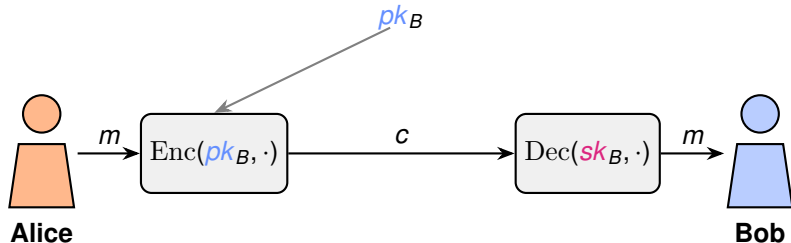
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what if encryption and decryption used *different* keys?
- Everyone has a key pair (pk , sk) and
 - publishes the **public key** pk ;
 - keeps the **private key** sk secret.
- Anyone can encrypt *to* you using pk ; only you can decrypt using sk .
- No prior interaction or shared secret needed!

Public Key Encryption (PKE)

Public Key Infrastructure (PKI)



KEM/DEM paradigm

- PKE is computationally expensive
- SKE has key distribution problems
- In practice: KEM/DEM paradigm

KEM: Key Encapsulation Mechanism

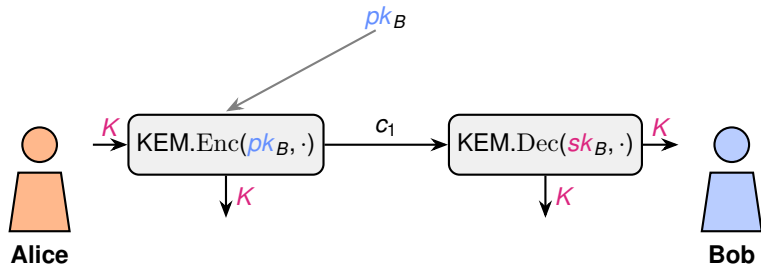
Use PKE to encrypt a short-lived symmetric key

DEM: Data Encapsulation mechanism

Use SKE to encrypt bulk data.

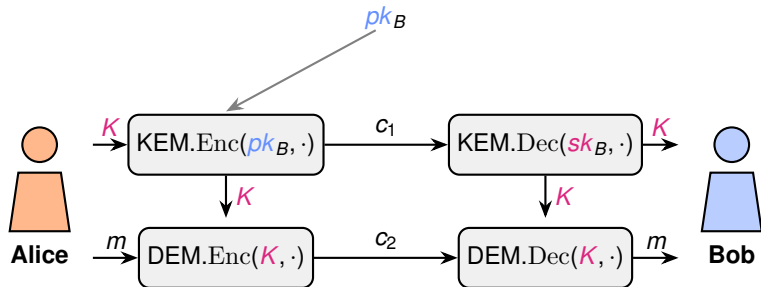
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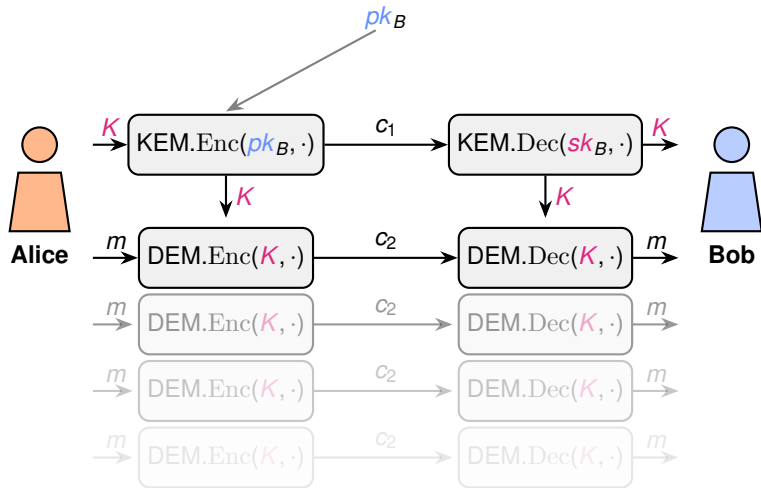
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PKE is everywhere

Communication

- **TLS/HTTPS**

key exchange when you
load `https://...`

- **Messaging**

Signal, WhatsApp(X3DH)

- **Email**

PGP, S/MIME

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- **Secure HW**

attestation, Secure
Enclave

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ballots & tallying

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Computation

- **Cloud (FHE)**

computation on cipher-
texts

- **Private PIR**

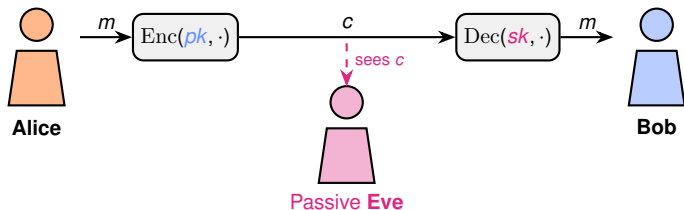
query w/o revealing what
you access

Passive vs. Active Adversary

- Intuitively: PKE “Hides the message”.

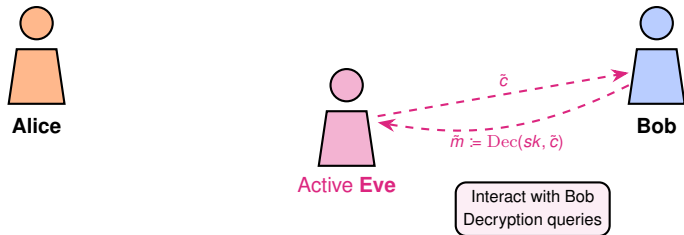
Passive vs. Active Adversary

- Intuitively: PKE “Hides the message”.
- But from whom?
 - Can the attacker only listen to the channel



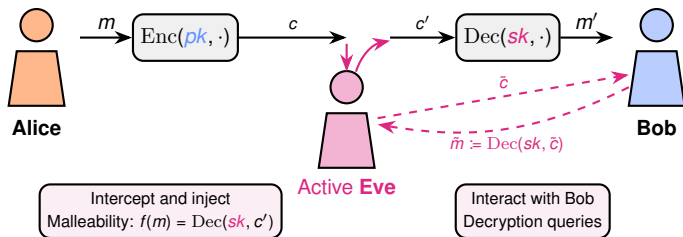
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 - and **inject**, **modify**, or **replay** ciphertexts.



First part of this course

Active adversaries (IND-CCA security)

But sometimes we *want* malleability!

- IND-CCA: “malleability of ciphertexts is BAD!”
- **BUT:** Malleability can also be a feature!
 - outsourcing computation to an untrusted cloud,
 - running machine-learning inference on encrypted data,
 - privacy-preserving statistics / e-voting (tallying encrypted votes),
 - private information retrieval.
- Goal: compute on ciphertexts *without decrypting*, and without leaking the underlying data.

Fully Homomorphic Encryption (FHE): a teaser

Homomorphic property:

$$\text{Enc}(m_1) \boxplus \text{Enc}(m_2) = \text{Enc}(m_1 \odot m_2)$$

for some operation $\odot$ on plaintexts, computed *directly* on ciphertexts.

- *Partially* homomorphic: support one operation (e.g. ElGamal: multiplication; Paillier: addition).
- *Fully* homomorphic (Gentry, 2009): arbitrary computation (both $+$ and $\times$) on encrypted data

Second part of this course

Computation of encrypted data (FHE, lattices)

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- Syntax and Formal Security Definitions (passive adversary)
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- Fully Homomorphic Encryption
 - Introduction to Lattice-based Cryptography
 - Step-by-step getting to FHE

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Semantic security equivalent to IND-CPA

Notation and Abbreviations

- $\mathbb{N}$ set of natural numbers, $\mathbb{Z}$ set of integers, $\mathbb{Z}_p$ set of integers modulo p
- Sampling of a random value:
 - $x \leftarrow \mathcal{D}$ sampling x according to the *distribution $\mathcal{D}$*
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 - $y := \mathcal{A}(x; r)$ output of a *probabilistic* algorithm $\mathcal{A}$ run on input x and random coins r .
- $y \in \mathcal{A}(x)$: there exists random coins r s.t. $y = \mathcal{A}(x; r)$

Notation and Abbreviations

- PPT = Probabilistic Polynomial Time algorithm
- $\lambda \in \mathbb{N}$ denotes the security parameter
- Negligible function: A function $\text{negl} : \mathbb{N} \rightarrow \mathbb{R}$ is *negligible* iff

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Examples: $1/2^k$ negligible in k , $1/k^2$ not.

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Definition 1.1: Public Key Encryption (PKE) Scheme - syntax

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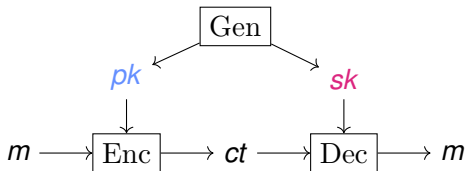
(for $m \in \mathcal{M}$)

- $\text{Dec}(sk, ct)$: outputs a message $m \in \mathcal{M}$ or an error $\perp$.

(intuitively: $\perp$ signals decryption failure.)

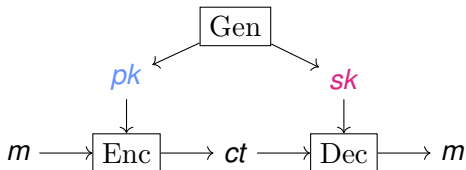
Correctness

Intuitively: The PKE scheme works.



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Definition 1.2: Perfect Correctness

$(\text{Gen}, \text{Enc}, \text{Dec})$ satisfies (perfect) correctness if

$$\forall \lambda \in \mathbb{N} \quad \forall (pk, sk) \in \text{Gen}(1^\lambda) \quad \forall m \in \mathcal{M} :$$

$$\Pr[\text{Dec}(sk, \text{Enc}(pk, m)) = m] = 1.$$

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What knowledge and power does the adversary have?
 - (Computational model)
E.g., Random Oracle Model

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Semantic Security – Intuition

No PPT adversary can learn any non-trivial information about m from ct .

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- ...but how to formalize this?
- Real/Ideal world paradigm (aka simulation-based security)

Real world

Describes what happens in a “real” situation.

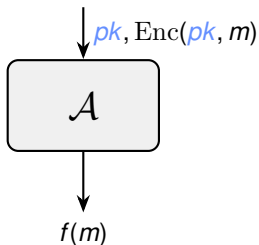
Ideal world

Describes how much leakage still okay.

Semantic Security – Intuition

- PPT adversary $\mathcal{A}$ gets $ct \leftarrow \text{Enc}(pk, m)$ for random m
- Has to output $f(m)$, i.e., “non-trivial information” about m

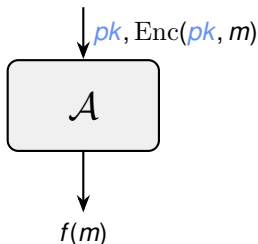
“Real” world



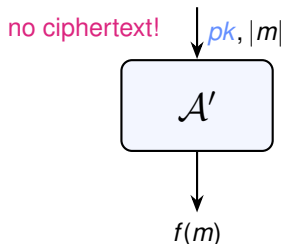
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- For every such $\mathcal{A}$ there exists PPT $\mathcal{A}'$ (called the *simulator*)
- ...which is equally likely to output $f(m)$ without seeing ct !

“Real” world



“Ideal” world



Semantic security

Random message?

- ...any efficiently sampleable distribution over $\mathcal{M}$

Not just uniform distribution!

- Formalization: PPT algorithm $(m, h) \leftarrow \text{Samp}(pk)$
 - m is the sampled message
 - h is some “*hint*”, leakage about m , public information

Semantic Security

Definition 1.3: Semantic security

A public key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is *semantically secure* if

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- **IND Security (INDistinguishability of encryption)**
 - “Game-based” definition for PPT adversaries
 - Focus of this course
 - Equivalent to semantic security

Game-based definition

- Security defined through an experiment (game)
- Experiment = interactive protocol between two parties
 - Adversary $\mathcal{A}$
 - Challenger Chal
- Adversary plays against the challenger *wins* if it reaches their goals.

IND-CPA security experiment

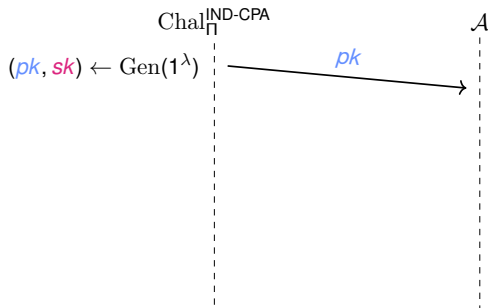
Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme.

Chal _{Π} ^{IND-CPA}

$\mathcal{A}$

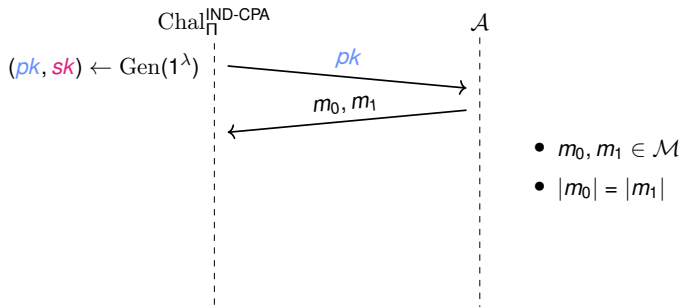
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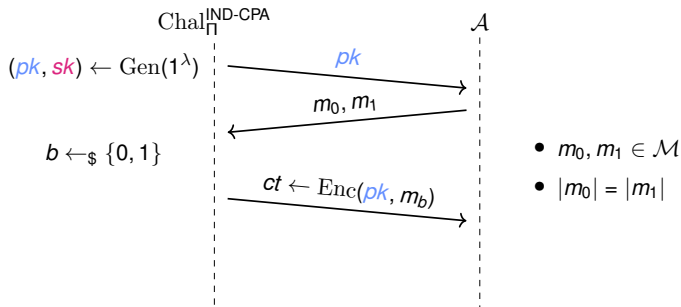
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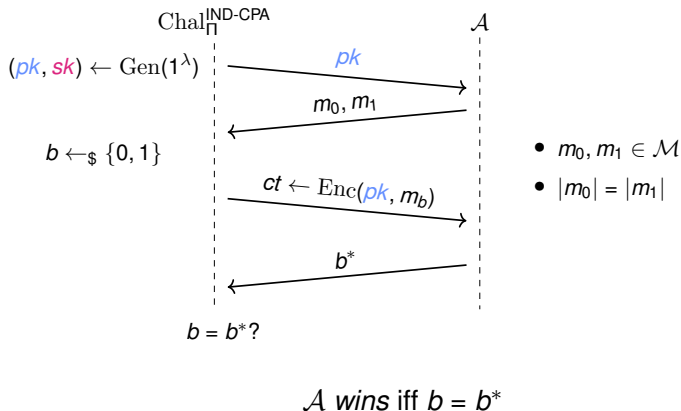
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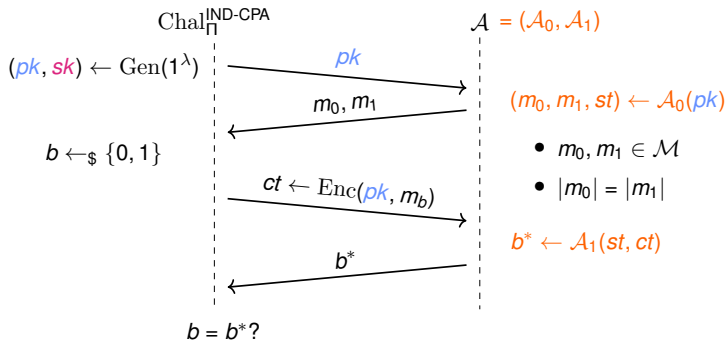
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$\mathcal{A}$ wins iff $b = b^*$

Indistinguishability under Chosen Plaintext Attack

Definition 1.4: IND-CPA security - asymptotic

A public key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is *indistinguishable under chosen plaintext attack* (IND-CPA) if for all PPT adversaries $\mathcal{A}$, the function

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CPA}}(\lambda) := 2 \cdot \left| \frac{1}{2} - \Pr[\mathcal{A} \text{ wins IND-CPA experiment}] \right|$$

is negligible in λ .

IND-CPA definition - concrete security

Sometimes it is useful to specify the *runtime* and the *success probability* of the adversary (not just say PPT and negligible).

Definition 1.5: [def:INDCPAconcrete]

IND-CPA security - concrete A public key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is (t, ϵ) -IND-CPA secure if for all PPT adversaries $\mathcal{A}$ running in time $\leq t = t(\lambda)$ the advantage is $\leq \epsilon = \epsilon(\lambda)$.

Alternative definitions of IND-CPA

- Goal above: distinguish encryption of two messages (aka) *“left-or-right”* IND-CPA
- Alternative: distinguish encryption of *“real-or-random”* message
 - $\mathcal{A}$ chooses $m_0 \in \mathcal{M}$, Chal chooses $m_1 \xleftarrow{\$} \mathcal{M}$
 - Chal returns $\text{Enc}(\textcolor{blue}{pk}, m_b)$
- Equivalent definitions (see exercise sheet)

Outline

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Game-based security

Semantic security equivalent to IND-CPA

Semantic Security $\Leftrightarrow$ IND-CPA

Theorem 1.1: Semantic Security $\Leftrightarrow$ IND-CPA

Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a public-key encryption scheme with message space $\mathcal{M}$. Π is IND-CPA secure if and only if it is semantically secure.

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We show each implication separately:

- Part I: Semantic Security $\Rightarrow$ IND-CPA
Easy proof by contrapositive

Semantic Security $\Leftrightarrow$ IND-CPA

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- Part I: Semantic Security $\Rightarrow$ IND-CPA
Easy proof by contrapositive
- Part II: IND-CPA $\Rightarrow$ Semantic Security
Slightly harder, hybrid argument

Proof (Part I): Semantic Security $\Rightarrow$ IND-CPA

Prove by contrapositive: $\neg \text{IND-CPA} \Rightarrow \neg \text{Semantic Security}$

- $\neg \text{IND-CPA}$:

We assume $\exists$ PPT $\mathcal{D}$ with non-negligible $\text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}$

Proof (Part I): Semantic Security $\Rightarrow$ IND-CPA

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- $\neg \text{IND-CPA}$:

We assume $\exists \text{ PPT } \mathcal{D}$ with non-negligible $\text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}$

- $\neg \text{Semantic Security}$:

We prove

$$\neg(\forall \text{ PPT } \mathcal{A} \quad \exists \text{ PPT } \mathcal{A}' \quad \forall \text{ PPT } \text{Samp} \quad \forall \text{ PT } f: |\epsilon_{\mathcal{A}} - \epsilon_{\mathcal{A}'}| \leq \text{negl}(\lambda))$$



$$\exists \text{ PPT } \mathcal{A} \quad \forall \text{ PPT } \mathcal{A}' \quad \exists \text{ PPT } \text{Samp} \quad \exists \text{ PT } f: |\epsilon_{\mathcal{A}} - \epsilon_{\mathcal{A}'}| > \text{negl}(\lambda)$$

Proof (Part I): Semantic Security $\Rightarrow$ IND-CPA

Assuming $\mathcal{D} = (\mathcal{D}_0, \mathcal{D}_1)$ breaks IND-CPA with probability $\epsilon_{\mathcal{D}}$.

We show:

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Samp(pk)

1. $(m_0, m_1, st) \leftarrow \mathcal{D}_0(pk)$
2. $b \leftarrow_{\$} \{0, 1\}$
3. Output $m := m_b$,
 $h := (m_0, m_1, st)$

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$f(m, h)$

1. Parse
 $h = (m_0, m_1, st)$
2. If $m = m_0$: return 0
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$\mathcal{A}(pk, ct, h)$

1. Parse
 $h = (m_0, m_1, st)$
2. Output
 $\tilde{b} \leftarrow \mathcal{D}_1(st, ct)$

Proof (Part I): Semantic Security $\Rightarrow$ IND-CPA

- If $\mathcal{D}$ wins, $\mathcal{A}$ wins, hence $\epsilon_{\mathcal{A}} = \epsilon_{\mathcal{D}}$
- Consider any PPT simulator $\mathcal{A}'$:
 - $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$
 - $(m := m_b, h := (m_0, m_1, st)) \leftarrow \text{Samp}(pk)$
 - $\mathcal{A}'$ gets $pk, h := (m_0, m_1, st), |m_b|$

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 - $(pk, sk) \leftarrow \text{Gen}(1^\lambda)$
 - $(m := m_b, h := (m_0, m_1, st)) \leftarrow \text{Samp}(pk)$
 - $\mathcal{A}'$ gets $pk, h := (m_0, m_1, st), |m_b|$
 - ▶ pk, h does not depend on the bit b
 - ▶ $|m_b|$ does not depend on the bit b (recall that $|m_0| = |m_1|$)
 - ▶ $\mathcal{A}'$ (information theoretically) *knows nothing about b*
 - ▶ Probability that $\mathcal{A}'$ wins $\epsilon_{\mathcal{A}'} = \frac{1}{2}$

Proof (Part I): Semantic Security $\Rightarrow$ IND-CPA

Putting things together:

- Assumption: $\text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}(\lambda)$ non-negligible
- We constructed $\text{Samp}, f, \mathcal{A}$ s.t. $\epsilon_{\mathcal{A}} = \epsilon_{\mathcal{D}}$
- We showed that for any $\mathcal{A}'$, we have $\epsilon_{\mathcal{A}'} = \frac{1}{2}$
- Hence:

$$\text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}(\lambda) \stackrel{\text{def.}}{=} 2 \cdot \left| \epsilon_{\mathcal{D}} - \frac{1}{2} \right| = 2 \cdot |\epsilon_{\mathcal{A}} - \epsilon_{\mathcal{A}}'|$$

- If $\text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}(\lambda)$ non-negligible, then also $\frac{1}{2} \text{Adv}_{\Pi, \mathcal{D}}^{\text{IND-CPA}}(\lambda)$ non-negligible.